

CARLISLE, United Kingdom

WMO: 032200

Lat: 54.9344N

Lon: 2.9636W

Elev: 89

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.1	27.3	21.8	16.4	29.4	24.1	18.3	31.5	35.3	46.7	32.1	47.7	1.0	80	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	74.8	64.0	71.4	62.3	68.4	60.7	65.8	72.5	63.8	69.3	62.1	66.9	5.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.3	87.4	68.2	61.5	81.9	66.3	60.0	77.6	64.7	30.6	72.7	29.2	69.3	27.9	67.1	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	23.3	19.9	20.3	82.2	3.6	4.3	17.7	85.3	15.6	87.8	13.6	90.3	10.9	93.4		
(5)			20.3	69.9	3.8	3.2	17.5	72.3	15.3	74.2	13.2	76.0	10.4	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	40.4	40.8	43.0	47.0	52.1	57.0	60.2	59.9	56.0	50.7	44.7	40.4	(6)
(7)		DBStd	8.90	5.80	5.34	4.90	4.78	4.93	4.07	3.95	3.76	4.10	5.12	5.96	7.10	(7)
(8)		HDD50	1459	300	260	222	112	32	1	0	0	4	53	175	300	(8)
(9)		HDD65	5716	764	678	683	539	401	243	159	167	269	442	610	761	(9)
(10)		CDD50	1237	1	2	4	23	97	210	316	305	185	76	15	3	(10)
(11)		CDD65	21	0	0	0	0	1	2	10	7	1	0	0	0	(11)
(12)		CDH74	151	0	0	0	0	6	26	72	43	4	0	0	0	(12)
(13)		CDH80	21	0	0	0	0	0	3	12	6	0	0	0	0	(13)
(14)	Wind	WSAvg	6.5	8.5	8.9	8.1	7.3	6.1	5.5	5.0	4.4	5.1	6.0	6.3	6.7	(14)
(15)	Precipitation	PrecAvg	30.6	2.8	2.0	2.2	1.9	2.1	2.3	2.7	3.0	3.1	3.1	2.9	2.8	(15)
(16)		PrecMax	40.3	5.3	6.9	4.4	4.4	4.2	6.5	7.4	6.2	7.0	8.4	6.9	5.1	(16)
(17)		PrecMin	22.5	0.2	0.2	0.7	0.2	0.7	0.2	0.8	0.2	0.5	0.6	0.9	0.5	(17)
(18)		PrecStd	4.4	1.3	1.4	0.9	0.9	0.9	1.3	1.3	1.5	1.4	1.5	1.4	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	55.3	59.1	67.3	73.9	77.3	81.0	79.6	72.4	64.8	59.2	56.3	(19)
(20)			MCWB	51.4	50.4	50.8	57.2	61.7	65.7	66.0	66.4	62.8	58.0	55.5	52.9	(20)
(21)		2%	DB	52.0	52.2	54.5	60.8	68.6	72.0	75.0	72.7	68.1	62.0	56.7	54.1	(21)
(22)			MCWB	49.3	49.0	49.5	52.5	58.8	62.0	64.5	63.4	60.6	57.1	53.7	51.4	(22)
(23)		5%	DB	50.3	50.7	52.2	57.5	64.2	67.9	71.3	69.0	65.3	60.2	54.9	52.0	(23)
(24)			MCWB	47.8	47.8	48.2	50.8	56.5	59.8	63.0	61.7	59.1	56.2	52.1	49.5	(24)
(25)		10%	DB	48.7	48.9	50.4	55.0	60.6	65.0	67.9	66.7	63.2	58.5	53.2	50.2	(25)
(26)			MCWB	46.4	46.3	46.9	49.4	53.7	58.1	60.9	60.4	57.8	55.0	50.7	47.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	51.8	53.1	57.4	63.7	67.2	69.9	67.5	65.3	60.1	56.4	53.8	(27)
(28)			MADB	53.8	53.6	57.4	64.0	71.6	74.3	74.8	75.2	70.2	62.5	58.5	55.8	(28)
(29)		2%	WB	50.1	50.0	50.7	53.9	59.8	63.6	66.2	65.0	62.4	58.4	54.3	51.9	(29)
(30)			MADB	51.8	51.8	53.4	58.9	67.3	70.2	73.2	70.7	66.2	61.0	56.1	53.8	(30)
(31)		5%	WB	48.3	48.3	49.1	52.0	57.2	61.0	63.8	63.2	60.3	56.9	52.7	50.0	(31)
(32)			MADB	50.0	50.4	51.6	56.5	62.7	66.6	69.7	67.6	63.6	59.9	54.5	51.9	(32)
(33)		10%	WB	46.6	46.5	47.5	50.2	55.1	58.9	61.9	61.7	58.7	55.1	50.9	48.1	(33)
(34)			MADB	48.5	48.8	50.1	54.3	59.7	63.7	66.7	65.5	62.2	57.9	52.9	50.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.8	11.3	13.4	14.4	13.7	13.5	13.1	12.8	11.2	9.6	9.2	(35)
(36)			MCDBR	9.2	10.3	13.0	17.3	20.1	19.6	19.7	17.6	15.9	11.7	9.1	9.8	(36)
(37)			MCWBR	8.6	8.4	9.0	11.0	12.1	11.8	11.0	10.1	9.7	8.5	8.3	9.2	(37)
(38)		5% WB	MCDBR	9.3	9.4	11.8	15.8	17.8	17.7	17.1	15.2	13.2	10.5	9.5	10.1	(38)
(39)			MCWBR	9.0	8.5	9.1	10.7	11.6	11.6	10.7	9.6	9.5	8.7	8.9	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.339	0.368	0.389	0.391	0.389	0.389	0.377	0.362	0.346	0.322	0.312	(40)	
(41)		taud	2.247	2.328	2.275	2.250	2.291	2.327	2.331	2.377	2.422	2.410	2.326	2.228	(41)	
(42)		Ebn at Noon	186	233	252	264	270	272	269	266	254	228	189	163	(42)	
(43)		Edh at Noon	19	25	32	38	38	37	37	33	28	23	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	174	382	725	1164	1516	1504	1452	1171	835	452	223	132	(44)	
(45)		RadStd	24	41	71	121	116	148	156	89	71	55	22	12	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page